

Why is the Newcastle disease virus still threatening the poultry industry despite vaccine implementation?

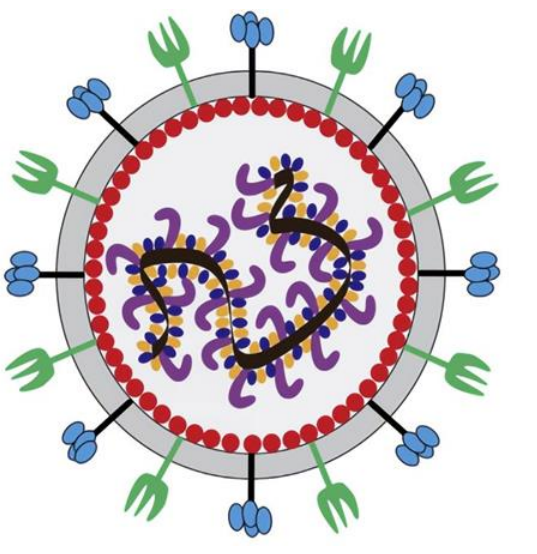
Development of a pseudotyped virus-based sero-neutralization assay for Newcastle Disease Virus: assessing vaccine efficacy against velogenic strains

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BACKGROUND

Newcastle disease (ND) is a severe, systemic and fatal viral disease of poultry due to virulent strains of avian paramyxovirus type 1, that attacks the respiratory, nervous and digestive systems¹.

ND vaccines do not provide sterile immunity.

Despite of large-scale vaccine implementation, ND has an important economic impact, affecting human welfare by decreasing food supplies².



- Is mass vaccination effective?
- Is there a risk of immune escape due to strain mismatches?

Vaccine strain	≠	Circulating strains
LaSota genotype II (low pathogenicity or lentogenic)		VII.2 subgenotype ⁴ (high pathogenicity or velogenic)

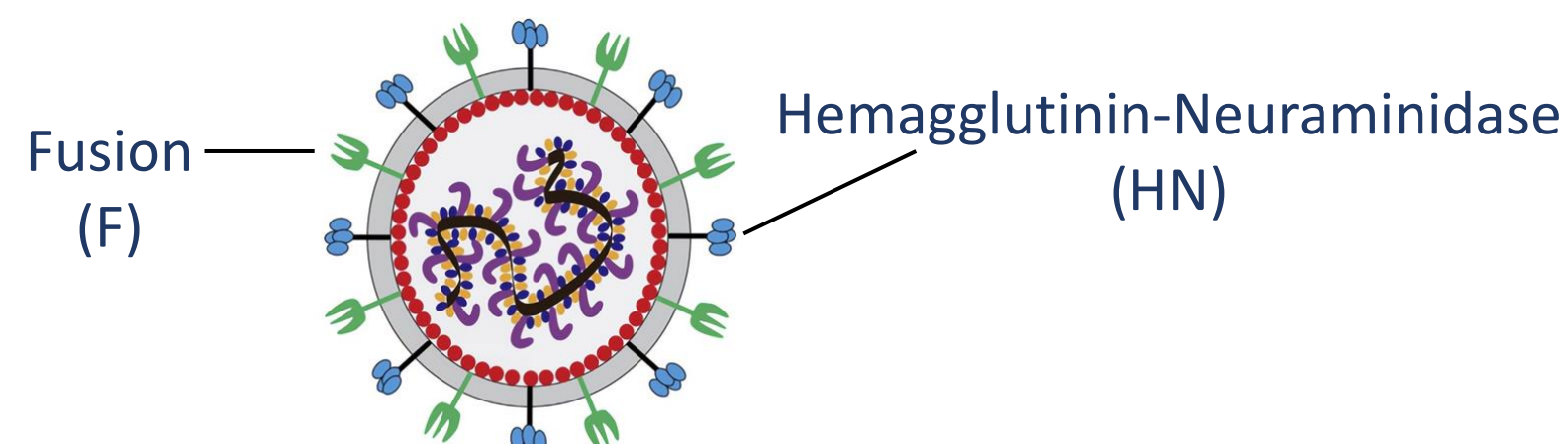
Measuring **functional neutralizing antibodies** could give information about **vaccine efficacy**.

PROBLEM

- Conventional sero-neutralization assays utilize live velogenic strains.
- This requires Biosafety Level (BSL) 3 handling.
- Safer alternatives are needed.

OBJECTIVE

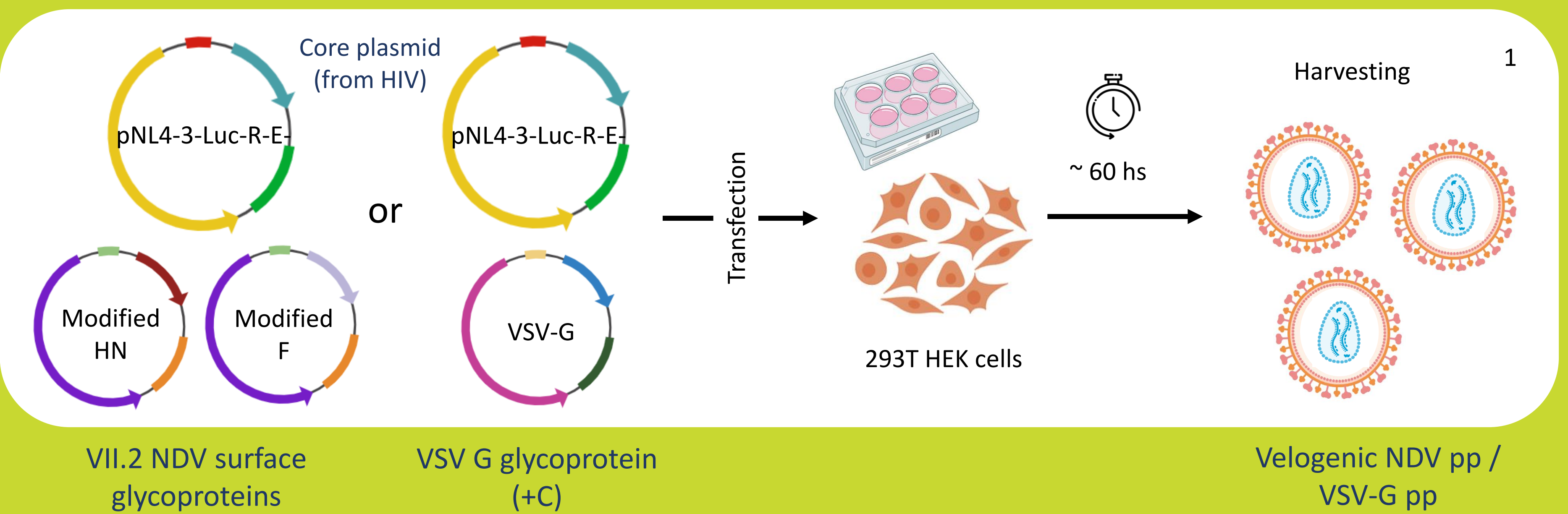
To develop a sero-neutralization assay employing pseudotyped viruses bearing the Newcastle Disease Virus (NDV) surface glycoproteins HN and F, the major antigenic determinants, which are crucial in the viral entry process.



This approach enables safer screening of extensive sera from vaccinated chickens in a BSL-2 laboratory.

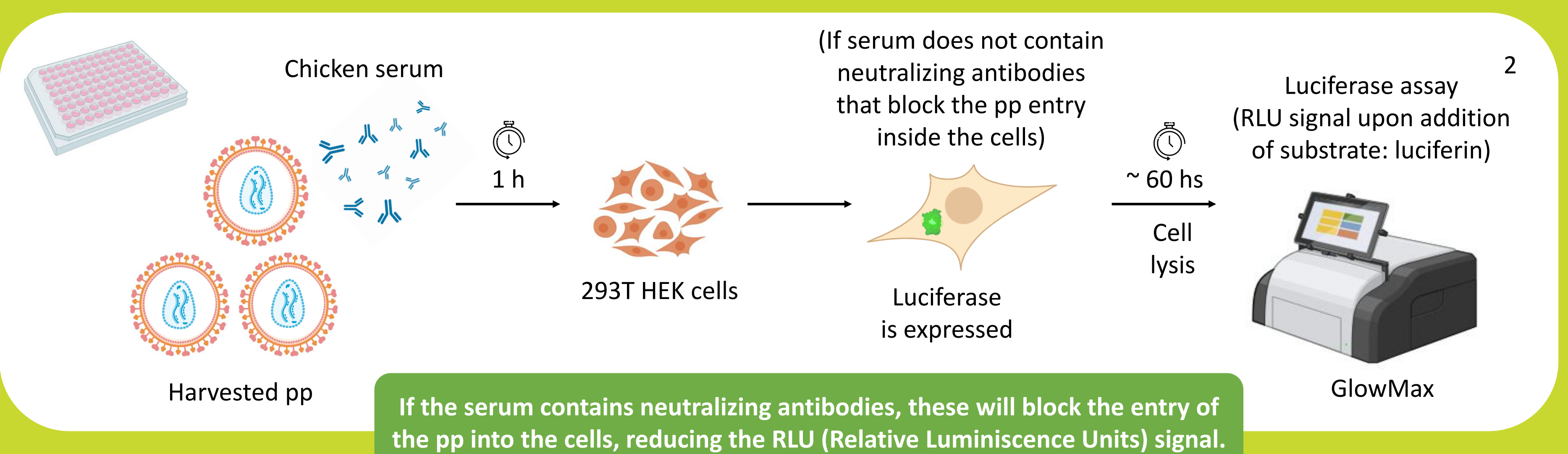
METHODOLOGY

Step 1: Pseudoparticle production



- Core plasmid pNL4-3-Luc-R-E- , it contains a reporter gene: the luciferase enzyme.
- Modified HN and F from NDV VII.2 subgenotype (2018 BeNeLux outbreak).
- VSV-G – Glycoprotein G from the Vesicular Stomatitis Virus (positive control)

Step 2: Sero-neutralization (SN) assay



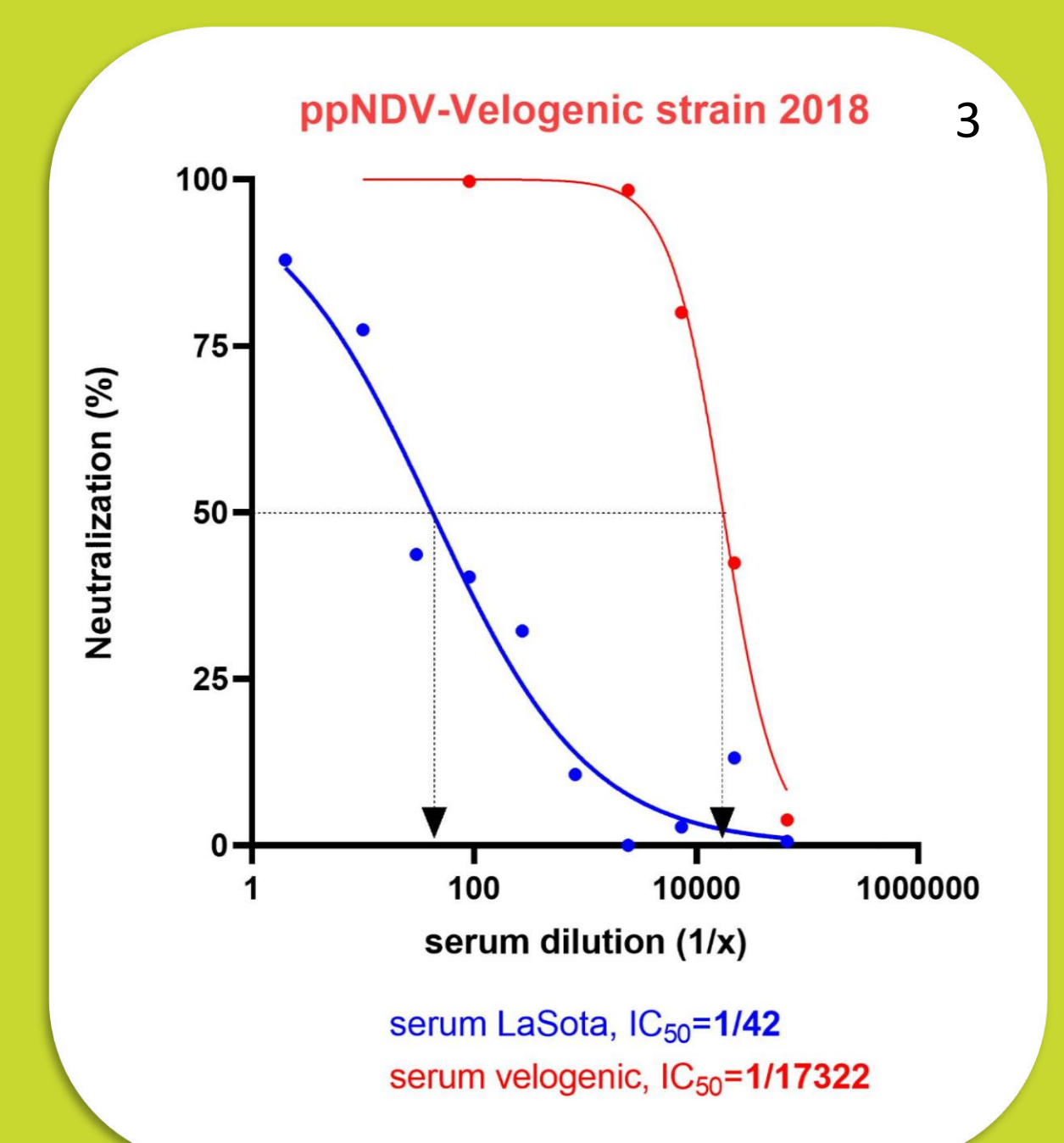
Two different reference sera were tested in the SN assay:

1. Pool of sera from chickens challenged with the live velogenic VII.2 NDV strain: velogenic reference serum.
2. Pool of sera from chickens challenged (vaccinated) with the LaSota strain: LaSota reference serum.

RESULTS

- Velogenic NDV pp and VSV-G pp (+C) were successfully produced.
- SN assays for LaSota and velogenic reference sera were performed and IC₅₀ values were calculated for each serum (Fig. 3).
- The velogenic serum (red) potentially neutralizes the velogenic NDVpp, but the LaSota serum (blue) can barely neutralize it.

The velogenic NDVpp can be neutralized by the velogenic serum, and not by the LaSota serum.



CONCLUSION

The pseudotyped virus-based SN assay promises to be a valuable tool for assessing *in vitro* vaccine efficacy against velogenic NDV strains, offering insights into the dynamics of the immune response and potential immune escape.

REFERENCES

- ¹ Newcastle Disease in Poultry (Avian Pneumoencephalitis, Exotic Newcastle Disease). <https://www.merckvetmanual.com/poultry/newcastle-disease-and-other-paramyxovirusinfections/newcastle-disease-in-poultry>
- ² Dimitrov, K. M., Afonso, C. L., Yu, Q., & Miller, P. J. (2017). Newcastle disease vaccines—A solved problem or a continuous challenge? *Veterinary microbiology*, 206, 126–136.
- ³ Preliminary Outbreak Assessment. Newcastle disease in Europe. 19th January 2023. <https://assets.publishing.service.gov.uk/media/63cea38ad3bf7f3c3fa85a10/newcastle-disease-europe-230123.pdf>
- ⁴ Steensels, M., Van Borm, S., Mertens, I., Houdart, P., Rauw, F., Roupie, V., ... & Lambrecht, B. (2021). Molecular and virological characterization of the first poultry outbreaks of Genotype VII. 2 velogenic avian orthomyxovirus type 1 (NDV) in North-West Europe, BeNeLux, 2018. *Transboundary and Emerging Diseases*, 68(4), 2147–2160.

FUTURE PERSPECTIVES

- To produce LaSota pp and to perform the SN assay using both LaSota and velogenic reference sera.
- To compare the pp-based SN assay with the live virus SN assay.
- To optimize the SN assay using the DF-1 (chicken embryo fibroblast) cell line.
- To evaluate the ability of the sera from vaccinated chickens of neutralizing the velogenic NDV pp.